

```

import pandas as pd

def gradient(w):
    return 4 * (w - 2)**3

def gradient_descent(w, learning_rate, iterations):
    weight_history = []
    weight_history.append(round(w, 4))
    for _ in range(iterations):
        grad = gradient(w)
        w = w - learning_rate * grad
        weight_history.append(round(w, 4))
    return weight_history

w0 = -1
iterations = 5
learning_rates = [0.01, 0.1, 0.5]

results = {}

for lr in learning_rates:
    results[lr] = gradient_descent(w0, lr, iterations)

df = pd.DataFrame(results)
df.index.name = 'Iteration'
df.index += 1
print("Weight values for each learning rate after 5 iterations:")
print(df)

```

Weight values for each learning rate after 5 iterations:

	0.01	0.10	0.50
Iteration			
1	-1.0000	-1.000000e+00	-1.000000e+00
2	0.0800	9.800000e+00	5.300000e+01
3	0.3631	-1.800208e+02	-2.652490e+05
4	0.5385	2.412074e+06	3.732511e+16

```
5      0.6634 -5.613462e+18 -1.040000e+50
6      0.7589  7.075424e+55  2.249727e+150
```

```
import pandas as pd
```

```
data = {
    0.01: ['Slow', 'No', 'No'],
    0.1: ['Moderate', 'Yes', 'Yes'],
    0.5: ['Fast', 'Yes', 'Yes']
}
```

```
df = pd.DataFrame(data, index=['Speed of Convergence', 'Oscillation', 'Overshoot'])
```

```
print(df)
```

	0.01	0.10	0.50
Speed of Convergence	Slow	Moderate	Fast
Oscillation	No	Yes	Yes
Overshoot	No	Yes	Yes

